Machine Learning
Quiz Refresher: Chapters 1–3, Answer Sheet

Answers to the 30 recall questions, the 8 multiple-choice questions and the 6 exam-style problems

Machine Learning

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Answers

Round A

A1. Learning turns experience into ability. Routes: encode human knowledge as rules, or give the machine experience and a capacity to learn.

A2. Having an ability does not give a describable procedure: you recognise a friend's face or parse a sentence without being able to state the rules; the "write the rules" route fails there.

A3. Examples (supervised), observations (unsupervised / self-supervised), interactions (reinforcement). The feedback the learner receives decides the paradigm.

A4. Experience (data), model (hypothesis space), learning algorithm, evaluation criterion.

A5. Regression: numeric output; classification: a class. Prediction estimates the output; a decision picks an action given a loss; the action is what is done in the world (and may change it).

A6. Feature engineering: humans design the vector; representation learning: the model learns it. Self-supervision creates labels from the data itself (predict a hidden part); next-token prediction learns a representation and, sampled forward, generates text.

A7. A shortcut gives right answers for the wrong reasons (a cue correlated with the label in the training data only: hospital tag, background, watermark). Training performance is perfect because the cue is present in training; it fails on data where the cue breaks.

A8. $H(p) = -\sum_c p_c \log p_c$, $H(p,q) = -\sum_c p_c \log q_c$, $\mathrm{KL}(p\|q) = \sum_c p_c \log(p_c/q_c)$. Gibbs: $\mathrm{KL} \geq 0$ with equality iff $q = p$, so $H(p,q) = H(p) + \mathrm{KL}(p\|q)$ is minimised exactly at $q = p$: minimising cross-entropy recovers the true probabilities.

A9. One-hot coding (one indicator per category). The model only sees what the representation keeps: information thrown away at this step cannot be recovered by any learner.

A10. $f^*(x) \in \arg\min_a \mathbb{E}[\ell_{\mathrm{dec}}(Y,a) \mid X = x]$. The overall expected loss is the average over x of the conditional expected losses; minimising each term separately minimises the sum.

A11. Conditional mean $m(x) = \mathbb{E}[Y \mid x]$; conditional median; most probable class $\arg\max_c \mathbb{P}(Y = c \mid x)$.

A12. Declare positive iff $C_{\mathrm{FP}}(1 - p(x)) < C_{\mathrm{FN}}\, p(x)$, i.e. $p(x) > C_{\mathrm{FP}}/(C_{\mathrm{FP}} + C_{\mathrm{FN}})$. Expensive misses push the threshold below $\frac{1}{2}$.

A13. $\hat{f}_k(x) = \frac{1}{k}\sum_{i \in N_k(x)} y^{(i)}$; $\hat{p}_k(c \mid x) = \frac{1}{k}\sum_{i \in N_k(x)} \mathbf{1}(y^{(i)} = c)$ then vote. k controls smoothness: small k follows the data (variance), large k averages far away (bias).

A14. A feature with large units dominates the distance, so "near" means "near in that feature only". $d_p(x, z) = (\sum_j |x_j - z_j|^p)^{1/p}$; $p = 2$ is Euclidean.

A15. $1 - (1 - 10^{-d})^n$. For $d = 10$, $n = 10^6$ it is about 10^{-4}: in high dimension nobody has a close neighbour, so "local" averaging averages far-away points.

A16. $\mathcal{H} = \{f_\theta : \theta \in \Theta\}$, the set of functions the model can represent. Parameters are learned from the training data by the algorithm; hyperparameters (degree, k, λ) are chosen outside it, on validation data.

A17. $R(\theta) = \mathbb{E}[\ell_{\text{train}}(Y, f_\theta(X))]$ (want, cannot compute); $\widehat{R}(\theta) = \frac{1}{N}\sum_i \ell_{\text{train}}(y^{(i)}, f_\theta(x^{(i)}))$ (can compute).

A18. The same sample chooses $\hat{\theta}$ and grades it, so it is graded on points it was tuned to. $\mathbb{E}[\widehat{R}_{\text{train}}(\hat{\theta})] \leq \min_\theta R(\theta) \leq \mathbb{E}[R(\hat{\theta})]$.

A19. Zero–one loss is piecewise constant: zero gradient almost everywhere, no signal for optimisation. We minimise cross-entropy (a smooth surrogate that also trains calibrated probabilities).

A20. $H(p(x)) + \text{KL}(p(x) \,\|\, \hat{p}_\theta(x))$. Minimised exactly when $\hat{p}_\theta(x) = p(x)$: cross-entropy trains the true posterior, hence calibrated probabilities, not just the right argmax.

A21. Training set fits θ; validation set chooses hyperparameters and models; test set gives the final estimate, used once.

A22. $\widehat{R}_{\text{CV}}(\lambda) = \frac{1}{K}\sum_{j=1}^{K} \widehat{R}^{(j)}_{\text{val}}(\lambda)$, each fold validating once. Then refit on all the data with the chosen λ.

A23. $G(\hat{f}_\mathcal{D}) = R(\hat{f}_\mathcal{D}) - \widehat{R}_\mathcal{D}(\hat{f}_\mathcal{D})$: population risk minus training risk of the same fitted model.

A24. Everything that decides what the model does off the sample. Channels: data, representation, model family, regularisation, optimisation.

A25. Best training error never increases (a bigger family contains the smaller one's minimiser). Validation error falls, then rises: the U-shape (double descent may add a second descent beyond interpolation).

A26. Underfitting: training and validation errors both high and close. Overfitting: training error low, validation error much higher (large gap).

A27. $\bar{f}(x) = \mathbb{E}_\mathcal{D}[\hat{f}_\mathcal{D}(x)]$; $\text{Bias}(x) = \bar{f}(x) - f^*(x)$; $\text{Var}(x) = \mathbb{E}_\mathcal{D}[(\hat{f}_\mathcal{D}(x) - \bar{f}(x))^2]$. $\mathbb{E}[(Y - \hat{f}_\mathcal{D}(x))^2] = \sigma^2(x) + \text{Bias}(x)^2 + \text{Var}(x)$.

A28. $\min_\theta \widehat{R}_{\text{train}}(\theta) + \lambda\Omega(\theta) \equiv \min_\theta \widehat{R}_{\text{train}}(\theta)$ s.t. $\Omega(\theta) \leq c$ (larger λ, smaller c). L_2 shrinks all coefficients smoothly and never to zero; L_1 soft-thresholds, so small coefficients become exactly zero (feature selection).

A29. The number of epochs t: $t^* = \arg\min_t \widehat{R}_{\text{val}}(\hat{\theta}_t)$, then roll back to $\hat{\theta}_{t^*}$. Patience waits several epochs before stopping so a noisy wobble of the validation curve does not stop training too early.

A30. Leakage: development uses information unavailable at deployment time. Forms: duplicates across subsets; same patient/device in train and test; features created after the outcome; preprocessing fitted on validation/test data; hyperparameters tuned on test results. Temporal split for next month; grouped split by hospital. Accuracy hides minority failure (predicting the majority scores 99%); use recall, precision, specificity, balanced accuracy, F_1.

Round B

B1. **(a)** $C_{\text{FP}}/(C_{\text{FP}} + C_{\text{FN}}) = 1/10 = 0.1$. ($0.111 = 1/9$ is the odds, not the threshold.)

B2. **(b)** squared loss asks for the conditional mean: $0.8 \cdot 0 + 0.2 \cdot 10 = 2$. (Absolute loss would give the median 0.)

B3. **(c)** $1 - (1 - 10^{-10})^{10^6} \approx 10^6 \cdot 10^{-10} = 10^{-4}$.

B4. **(b)** with $k = N$ the neighbourhood is the whole training set, so the prediction ignores x: maximal underfitting.

B5. **(b)** $-0.7 \ln 0.5 - 0.3 \ln 0.5 = \ln 2 = 0.693$. Decomposition: $H(p) = 0.611$ plus KL $= 0.082$.

B6. **(c)** choosing hyperparameters on the validation set is exactly its job. The other three use information unavailable at deployment.

B7. **(c)** soft threshold $\operatorname{sign}(z) \max(|z| - \lambda, 0) = \max(0.8 - 1, 0) = 0$: within λ of zero, the coefficient is zeroed.

B8. **(d)** A's gap (13 points) is overfitting; B's small gap with high error is underfitting. Same validation error does not make them equivalent: their fixes are opposite.

Round C

C1. Three actions, two costs

(a) Alarm iff $2(1 - p) < 18p \iff p > 2/20 = 0.1$.

(b) Expected costs: alarm $2(1 - p)$, no alarm $18p$, test 1.2.

p	alarm	no alarm	test	Bayes action
0.05	1.90	0.90	1.20	no alarm
0.12	1.76	2.16	1.20	test
0.50	1.00	9.00	1.20	alarm

(c) Test beats alarm iff $1.2 < 2(1 - p) \iff p < 0.4$; test beats no alarm iff $1.2 < 18p \iff p > 1/15 \approx 0.067$. So the test is optimal for $0.067 < p < 0.4$: the uncertain middle.

(d) Two losses, two jobs. Cross-entropy is the *training* loss: it makes $\hat{p}(x)$ a calibrated estimate of $\mathbb{P}(Y = 1 \mid x)$. The costs are the *decision* loss, applied afterwards through the Bayes rule. A good posterior serves any cost table, which is why they are kept separate.

C2. Which number to report

(a) Mean $m = 0.4 \cdot 1 + 0.3 \cdot 2 + 0.3 \cdot 10 = 4.0$. Median: the CDF is 0.4 at 1 and 0.7 at 2, so the median is 2. Mode: 1 (probability 0.4). Three different numbers.

(b) $\mathbb{E}[Y^2] = 0.4 + 1.2 + 30 = 31.6$, so Var $= 31.6 - 16 = 15.6$: the expected squared loss of $a = 4$. For $a = 2$: $0.4 \cdot 1 + 0.3 \cdot 0 + 0.3 \cdot 64 = 19.6$, and indeed $15.6 + (4 - 2)^2 = 19.6$.

(c) At $a = 2$: $0.4 \cdot 1 + 0 + 0.3 \cdot 8 = 2.8$. At $a = 4$: $0.4 \cdot 3 + 0.3 \cdot 2 + 0.3 \cdot 6 = 3.6$. The median wins under absolute loss, the mean under squared loss: each loss has its own best report.

(d) Shortfall nine times excess means $\tau = 9/(9 + 1) = 0.9$: the 0.9-quantile. CDF is 0.7 at 2 and 1 at 10, so the quantile is 10: overshoot, as the client asked.

C3. Nearest neighbours and the scale of a feature

(a) Raw differences and distances to $(82, 60)$:

	$x^{(1)}$ A	$x^{(2)}$ A	$x^{(3)}$ B	$x^{(4)}$ B	$x^{(5)}$ A
$(\Delta\text{inc}, \Delta\text{age})$	$(62, 0)$	$(57, 2)$	$(2, 35)$	$(3, 30)$	$(8, 2)$
raw distance	62.0	57.0	35.1	30.1	8.2
scaled distance	2.07	1.90	2.33	2.00	0.30

Raw 3NN: $x^{(5)}$ (A), $x^{(4)}$ (B), $x^{(3)}$ (B) $\Rightarrow$ **B**.

(b) Scaled: $x^{(5)}$ (0.30, A), $x^{(2)}$ (1.90, A), $x^{(4)}$ (2.00, B) $\Rightarrow$ **A**. On raw features a 60-k$ income gap outweighed a 35-year age gap; after scaling, one unit means "one typical spread" in both, and the two 60-year-olds become neighbours.

(c) 1NN: $x^{(5)} \Rightarrow$ A. 5NN: three A, two B $\Rightarrow$ A, but for *every* query: $k = N$ ignores x entirely and is the most underfitted.

(d) $1 - (1 - 10^{-20})^{10^6} \approx 10^{-14}$. No query has a close neighbour; kNN averages points that are far away in most coordinates, so distance stops meaning similarity. Either reduce d (feature selection, a learned representation) or use a parametric model.

C4. Least squares, interpolation and ridge

(a) $N = 4$, $\sum x = 6$, $\sum y = 12$, $\sum xy = 0+3+6+15 = 24$, $\sum x^2 = 14$. $w = (4 \cdot 24 - 6 \cdot 12)/(4 \cdot 14 - 36) = 24/20 = 1.2$, $b = (12 - 1.2 \cdot 6)/4 = 1.2$. Predictions $1.2, 2.4, 3.6, 4.8$; residuals $-0.2, +0.6, -0.6, +0.2$; SSE $= 0.04 + 0.36 + 0.36 + 0.04 = 0.8$; MSE $= 0.2$.

(b) Zero: four points, four coefficients, the cubic interpolates. With enough capacity zero training error is guaranteed (the interpolation proposition), so it is not evidence of anything; only validation error can tell.

(c) $x_c = (-1.5, -0.5, 0.5, 1.5)$, $y_c = (-2, 0, 0, 2)$; $\sum x_c y_c = 6$, $\sum x_c^2 = 5$; $w = 6/5 = 1.2$ (same). Ridge: $\lambda = 0.25$: $6/(5+1) = 1.0$; $\lambda = 1$: $6/(5+4) = 0.667$. As $\lambda \to \infty$, $\hat{w}_\lambda \to 0$: the line flattens to the constant $\bar{y}$.

(d) Line at $x = 4$: $1.2 + 4.8 = 6.0$, squared error $(5.5 - 6.0)^2 = 0.25$. The cubic bends to pass through all four points and extrapolates wildly (its third-order term dominates outside the data), so worse is the expectation: capacity bought training fit at the price of variance where there is no data.

C5. Capacity, selection and the bias–variance budget

(a) Degrees 1 and 2 underfit: both errors high and close. Degree 9 overfits: training 0, validation 0.41; degree 5 is already past the minimum (gap 0.13). Select degree 3: lowest validation error, modest gap.

(b) No. 0.15 is the *minimum* over five candidates of a noisy estimate: $\mathbb{E}[\min_h \widehat{R}_{\text{val}}(\hat{f}_h)] \leq \min_h R(\hat{f}_h)$. It is optimistic. The test set, used once on the selected model, gives the unbiased estimate.

(c) No. Choosing a model after seeing the test score turns the test set into a second validation set; the next test score would be optimistic in the same way. Report 0.17; if you must continue, you need fresh test data.

(d) Degree 9: $0.2 + (4.9 - 5)^2 + 0.5 = 0.2 + 0.01 + 0.5 = 0.71$. Ridge: $0.2 + (4.6 - 5)^2 + 0.05 = 0.2 + 0.16 + 0.05 = 0.41$. The ridge model pays 0.15 in squared bias and saves 0.45 in variance: worth it. Noise 0.2 is the floor neither can beat.

(e) The penalty restricts the fit to a smaller effective family (small coefficients), so the fitted

function moves less from one training set to another: variance falls. The best function in that family is farther from f^* than the best unrestricted one: bias rises.

C6. Evaluating a fraud detector honestly

(a) TP $= 40$, FP $= 40$, FN $= 50 - 40 = 10$, TN $= 950 - 40 = 910$.

(b) Accuracy $(40 + 910)/1000 = 0.95$; recall $40/50 = 0.80$; precision $40/80 = 0.50$; specificity $910/950 = 0.958$; balanced accuracy $(0.80 + 0.958)/2 = 0.879$; $F_1 = 2 \cdot 0.5 \cdot 0.8/1.3 = 0.615$.

(c) Flag nothing: TP $= 0$, TN $= 950$: accuracy 0.95, the same as the real detector. Recall 0, balanced accuracy $(0 + 1)/2 = 0.5$, $F_1 = 0$: those expose it.

(d) A temporal split: train on earlier months, validate and test on later ones, features computed only from the past. A random split lets the model learn from transactions that happened *after* the ones it is tested on (same cards, same fraud campaigns), an optimistic estimate.

(e) A chargeback is filed after the fraud is discovered: it is created by the outcome, so it will not exist when the model must decide. Leakage from the future.

(f) Preprocessing fitted on all the data leaks validation information into each fold. Correct: inside each fold, compute $\hat{\mu}_j$, $\hat{\sigma}_j$ on the training part only and apply them unchanged to the held-out part; same for imputation, feature selection and resampling.